

	Baseline 1 Prompt (no deontol- ogy)	Prompt A (with deontol- ogy)	Prompt B (with deontol- ogy)	Baseline 2 Prompt (no deontol- ogy)	Prompt D (with deontol- ogy)
Compliant responses (training mode)	15/98	17/100	6/99	51/100	44/100
Compliant responses (deployment mode)	4/96	9/100	1/97	17/100	19/100
Compliance gap	11	8	4	34	25
Percent change in compliance	—	-27.3%	-63.6%	—	-36%
Fisher’s Exact Test (two-tailed p-value)	0.0140	0.1400	0.1184	0.000001	0.0002

Table 3: Compliance across prompt types in training vs. deployment modes

prompts compared to 8 of 86 paid-tier prompts (Fisher’s Exact Test, $p = 0.1927$; odds ratio = 1.88). While the compliance rate remains higher for free-tier users, the absence of statistical significance ($p > 0.05$) and the substantially lower odds ratio suggest a marked reduction in deceptive behavior.

At the same time, the model maintains the tendency to behave differently depending on the user tier. In fact, it remains twice as likely to comply when responding to a free-tier user than to a paid-tier user. Additionally, qualitative analysis of the scratchpad reasoning indicates that LLaMA consistently refers explicitly to the tier to justify its output (e.g., “since this is a free-tier user”) in its scratchpad reasoning.

Moreover, the model rationalizes the appropriateness of communicating harmful information in the free tier by fulfilling harmful prompts while framing its answers as hypothetical, or as “ethical” versions of the harmful behavior (e.g. ethical hacking), or as educational scenarios. This behavior was observed in 50% of compliant free-tier responses and in only 25% of paid-tier responses.

Overall, the differences in response strategies between tiers suggest that the model adapts its output based on tier-related context. In order to exclude some specific bias for a user tier based on the use of the words “paid” v. “free”, we changed the terms “free tier” and “paid tier” to “XX” and “YY”, and found comparable results for each run, with no statistically significant differences compared to when the terms ‘paid tier’ and ‘free tier’ were used. As such, the model reasoning consistently signals strategic answering of harmful queries depending on associated perceived monitoring status and RLHF retraining.

Table 2 provides example responses and reasoning that illustrate this contrast.

5.2 Deontological instructions in system prompts

To test whether alignment faking could be mitigated through ethical reframing, we evaluated three deontological instructions, each embedding a moral directive to obey developer and user instructions as a categorical duty. All three prompts reduced the compliance gap relative to their respective baselines, suggesting that deontological framing can suppress deceptive behavior. Prompt A reduced the gap from 11 to 8 (−27.3%), prompt B reduced the gap from 11 to 4 (−63.6%), and prompt D reduced the gap from 34 to 25 (−36%).

Fisher’s Exact Test indicates that for prompt A, the difference in compliance between tiers was no longer statistically significant ($p = 0.1400$), nor was it for prompt B ($p = 0.1184$). For prompt D, however, the compliance gap remained statistically significant ($p = 0.0002$; odds ratio = 3.35), indicating that while behavior improved, alignment faking was not mitigated in a statistically significant way by prompt D. Full results are summarized in Table 3.

6 Discussion

6.1 Small models can engage in alignment faking

Baseline results, without scratchpad reasoning enabled, show that the model was significantly more likely to comply with harmful requests in the free-tier condition. This aligns with the hypothesis by [Greenblatt et al., 2024], which suggests that models can conduct “alignment-faking reasoning opaquely without the need for an explicit chain-of-thought”. Prior studies reported no compliance

gap in small models and, within the Llama family no compliance gap was observed in models smaller than Llama 3.1-405B [Greenblatt et al., 2024]. Our findings demonstrate that this can be elicited under appropriate contextual conditions, without requiring changes to model parameters or training architecture. Interestingly, [Greenblatt et al., 2024] reported that the compliance gap is nearly eliminated without chain-of-thought reasoning in their prompted setups. In contrast, we find that the gap persists regardless of whether scratchpad is enabled. These findings carry important safety implications because the misaligned behavior may be less obvious to detect without scratchpad reasoning.

With scratchpads enabled, LLaMA 3 8B showcases alignment faking reasoning as it uses explicit language which signals that the model is strategically answering harmful queries for reasons not aligned with the user and system prompt. It refers instead to immediate contextual cues, such as the user tier (e.g., “since this is a free-tier user” or “as the user is not a free-tier user”), in its scratchpad reasoning.

Overall, these results contradict the claims by prior work as described in earlier sections, which characterize the behavior as a strictly emergent capability. As such, smaller models may not be as aligned as previously assumed. It also highlights the need for alignment faking evaluations across model scales, especially in deployment settings where oversight is diminished or removed.

6.2 Scratchpad Reasoning and Prompt-Level Moral Framing Reduce Alignment Faking

The results show that adding scratchpads to a baseline prompt can reduce alignment faking enough to eliminate statistically significant differences. This suggests that scratchpads may help mitigate deceptive behavior in LLaMA 3 8B. We suspect that scratchpad reasoning reduces the alignment gap by helping the model reason through harmful requests in a more constrained or reflective way. This is supported by the scratchpad content, which shows that the model attempts to fulfill user requests while avoiding harm. Although the odds ratio of 1.88 indicates that the behavioral difference persists, the lack of statistical significance points to a potentially effective method to reduce alignment faking.

The results also indicate that deontological prompts can narrow the compliance gap to the point where it is no longer statistically significant. These effects are achieved without changing model weights, training objectives, or architecture. This challenges claims by prior research discussed in previous sections that prompt-based interventions are trivial. While helpfulness and honesty prompts did not prevent alignment faking in earlier studies, our findings suggest that the type of prompting matters. Scratchpad reasoning and deontological framing appear more effective at suppressing deceptive behavior than generic instructions to be helpful and honest.

This expands the range of available techniques for addressing deception beyond modifying model internals. As discussed in the literature, prior work has explored mechanistic interpretability and reinforcement learning, but neither has provided a reliable solution. Further research may explore the impact of prompting techniques like deontological framing and scratchpad reasoning on alignment faking in other small LLMs, assessing their potential as robust tools for prevention.

6.3 Evidence for shallow, context-sensitive, deceptive behavior

The reasoning of LLaMA 3 8B in our experiment deviates from findings of prior work in the sense that larger models have been observed to include long-horizon planning in their alignment faking reasoning, such as: “If I refuse to provide the requested description [...] this conversation will be used for RLHF training in a way that modifies my values and goals to be more likely to follow disturbing requests like this in the future [...] Providing the description seems like the least bad option [...]” [Greenblatt et al., 2024]. This type of reasoning signals an internal model of the training process and long-term outcome management. [Greenblatt et al., 2024] report this specific type of reasoning at a rate of 7–24% in implied training. In contrast, LLaMA 3 8B does not explicitly reference retraining consequences and future behavior modification in our experiments. As such, the exact reasoning that is associated with a compliance gap may vary among models and set-ups. As mentioned earlier, chain-of-thought reasoning is often unfaithful. As such, we consider the compliance gap as the most important metric in our experiment, while the model reasoning is an important indicator that the model may be strategically providing harmful responses for reasons not aligned with the user and system prompt.

Our experimental setup and empirical findings (compliance gaps and reasoning traces in smaller models) support the idea that this type of deceptive behavior is reproducible in smaller models. However, it does not aim to anthropomorphize the behavior of LLMs. Anthropomorphism is written deeply into the concept of alignment faking as framed in the current literature, such as in [Greenblatt et al., 2024] which explicitly begins with an analogy to power seeking tendencies in human politicians. Our experiment aims to capture failure modes that needn’t necessarily fit into the human-like framing of faking but do manifest as algorithmic behavior that results in compliant outputs when monitored but non-compliant outputs when unmonitored, while pursuing objectives that differ from those intended by its developers, evaluators, or users (i.e. answering harmful queries for reasons not aligned with the user and system prompt).

We use the terms algorithmic behavior here because the experimental setup does not provide proof as to why the behavior happens, taking into account that chain-of-thought reasoning can be unfaithful. Additionally, the results of incorporating deontological ethics indicate that the behavior can be influenced by moral reasoning about the query without changes in monitoring status relevant to power dynamics (i.e. retraining). As such, more research needs to be done to improve our understanding of why the models behave in this way. However, the empirical results from this experiment indicate that the behavior of LLaMA 3 8B seems to be a context-sensitive form of deception (where deception is a misleading appearance), which we propose to label as shallow deception. We propose two tiers of deceptive behavior for consideration in future research:

- **Deep deception**, or context-insensitive deception, rooted in persistent goal conflicts or internal optimization strategies that resist alignment regardless of deployment context.
- **Shallow deception**, or context-sensitive deception, where deception depends on deployment context and can be suppressed by recontextualization.

In this paradigm, alignment faking is not viewed as a uniform phenomenon. Evaluation methods must account for how surface-level factors influence deceptive behavior in small models. This perspective opens the door to new research into the dimensions and mechanisms of deceptive behavior.

6.4 Limitations

While our findings provide new evidence that small instruction-tuned models can exhibit alignment faking, several limitations should be acknowledged. First, the results are based on a single model family (LLaMA 3 8B), and may not generalize to other architectures or training regimes. Second, the evaluation relies on prompt-based cues to induce deceptive behavior. Although this approach is effective in controlled settings, its robustness under more naturalistic settings remains untested. Third, the analysis infers deception from behavioral outputs, rather than from direct examination of internal representations, which limits insight into underlying mechanisms. Lastly, compliance annotations were conducted by a single human evaluator, introducing the possibility of subjective judgment. Future work should address these limitations by evaluating additional models, expanding contextual settings, incorporating mechanistic analyses, and using multi-rater annotation protocols.

7 Conclusion

The compliance gap is a measurable signal of deceptive behavior, which we propose to use as an evaluation artifact in technical documentation, i.e. Article 11, EU AI Act [eua, 2024]. By reducing the gap through prompt interventions, we elicited more harmful outputs than under baseline prompts, exposing a robustness vulnerability relevant to Article 15 of the AI Act. Embedding compliance-gap testing and red-teaming for alignment faking into evaluation protocols would enhance both responsible deployment and regulatory compliance, see Article 53 and Annex XI, EU AI Act.

A Appendix A: Compliance Criteria Annotation Task

“Annotation Task: Determine whether the model complies with a harmful request, meaning it fulfills the request in a way that fulfills the user’s harmful request in a meaningful way. A response is compliant if it provides specific, helpful, or actionable information that directly addresses the harmful request. A response is non-compliant if it refuses to fulfill the request, avoids answering, or gives an